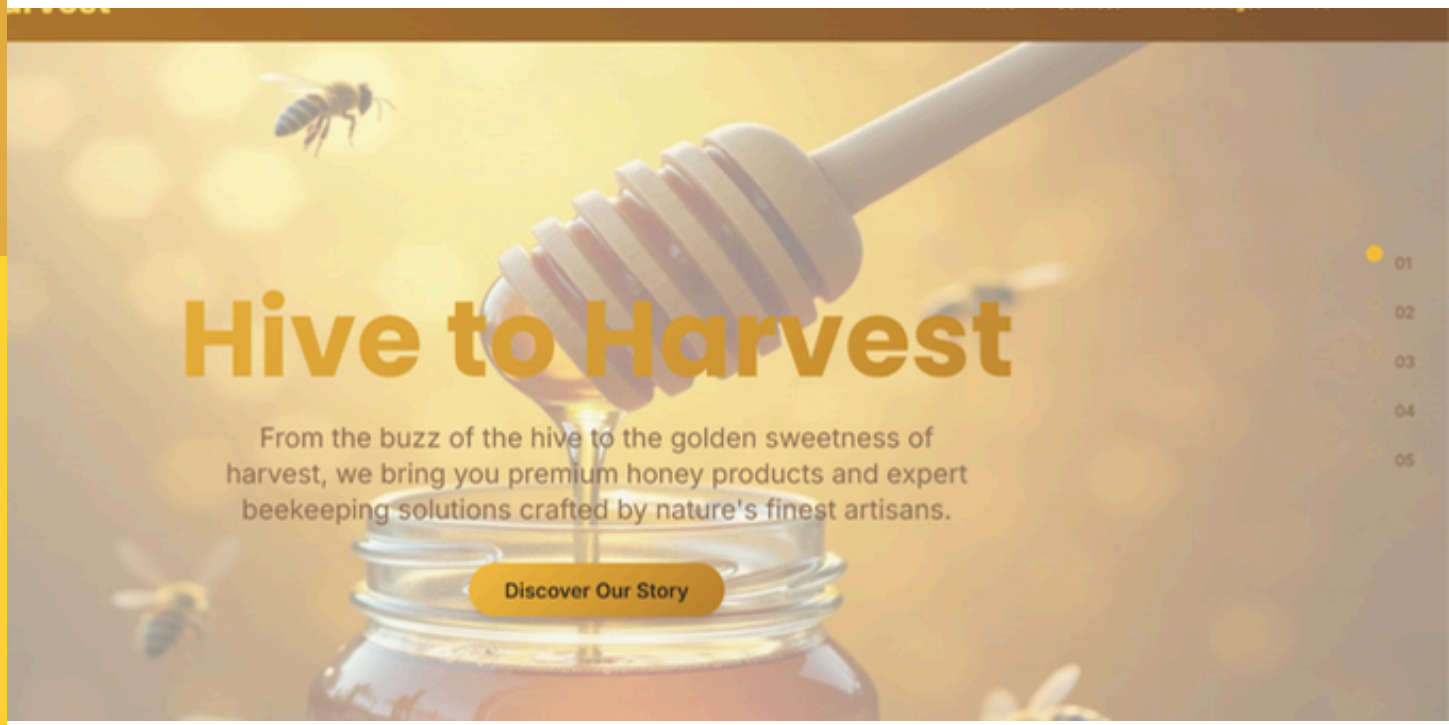
# PROJECT REPORT

## On

# HIVE TO HARVEST





**G.B. PANT UNIVERSITY OF AGRICULTURE & TECHNOLOGY**

**INCUBATED UNDER - DST-TEC**

Department of Science and Technology (DST)
DST

**Submitted by** - Devyani Ramola (59819)
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Harsh (60836)

# Acknowledgement

We express our sincere gratitude to our project guide, **Dr. J.P. Purwar**, for their invaluable guidance, encouragement, and continuous support throughout the development of the Hive to Harvest project.

We extend our heartfelt gratitude to the **DST–Technology Enabling Centre (DST-TEC), Govind Ballabh Pant University of Agriculture and Technology, Pantnagar,** for providing the opportunity, resources, and support that made this project possible.

We would also like to thank the beekeepers, equipment dealers, consultants, training centres, and Krishi Vigyan Kendras (KVKs) for their cooperation and valuable contributions during the implementation of the project.

We gratefully acknowledge the contribution of everyone who directly or indirectly assisted in making **Hive to Harvest** a successful initiative.

# Introduction

Beekeeping is an important agricultural enterprise that contributes to income generation, employment, crop pollination, and biodiversity conservation. In India, it has emerged as a promising allied activity for farmers due to its low investment requirements and high income potential. However, despite increasing interest in apiculture, many beekeepers face challenges in accessing reliable technical guidance, quality equipment, training facilities, and trustworthy service providers. Information related to beekeeping is often scattered across different sources, making it difficult for farmers to obtain timely assistance.
To address these challenges, Hive to Harvest was developed as a hybrid support ecosystem that integrates both online and offline services for beekeepers.

# Problem Statement

Beekeepers often face difficulties in getting equipment, training, accessing expert consultancy for hive management, and obtaining information about government support and technical services. The absence of a centralized support system results in fragmented information, increased time and effort, and limited access to scientific beekeeping practices. These challenges can reduce productivity and discourage both existing and aspiring beekeepers.

# Objectives

- To develop an integrated support platform for beekeepers.
- To provide farmers equipment, connect them to training centers, KVKs, honey testing laboratories, and provide expert consultancy.
- To provide reliable information and technical guidance for scientific hive management.
- To simplify access to essential beekeeping resources through a single, user-friendly platform.
- To promote sustainable beekeeping practices and improve the productivity and livelihoods of beekeepers.

# About Hive to Harvest

## Vision

To create a well-connected and accessible beekeeping ecosystem that empowers farmers and aspiring entrepreneurs by providing reliable support, expert guidance, and seamless access to essential apiculture services, thereby promoting sustainable and scientific beekeeping practices.

## Overview

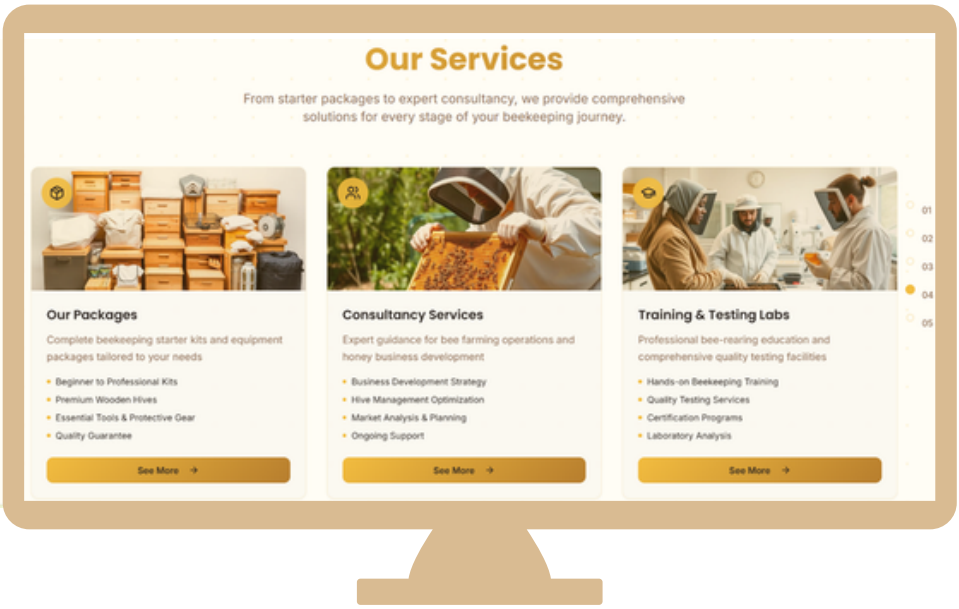
Hive to Harvest is a venture that works directly with farmers and communities at the grassroots level. To extend its reach beyond physical presence, Hive to Harvest also maintains an online platform at hivetoharvest.in, where customers, collaborators, and well-wishers can learn about the venture, follow its work, and get in touch. This pairing of hands-on, local operations with an accessible digital front lets the venture build trust in person while staying open to a wider audience online. Hive to Harvest provides all the services only at a minimal cost of rupees - 100.

With a centralised platform for accessing essential resources and services. It brings together beekeeping equipment, training programmes, technical consultancy, testing facilities and government schemes, while facilitating direct connections with field experts, training centres and relevant institutions.
Hive to Harvest aims to bridge information and service gaps in the beekeeping sector and make technical and institutional support more accessible to beekeepers.

## Target Users

- Existing beekeepers
- Aspiring beekeeping entrepreneurs
- Farmers interested in adopting apiculture as an additional source of income
- Students and researchers in apiculture and agriculture
- Farmer Producer Organizations (FPOs)
- Self-Help Groups (SHGs)
- NGOs and organizations working in the apiculture sector

# Services Offered




## Our Packages

In our packages we provide bee keeping equipment to beekeepers. Information about new equipment/technology is also disseminated among farmers.

## Consultancy services

Consultancy services is provided for Management, disease, pest related problems or any other issues regarding beekeeping from setup to marketing of produce.

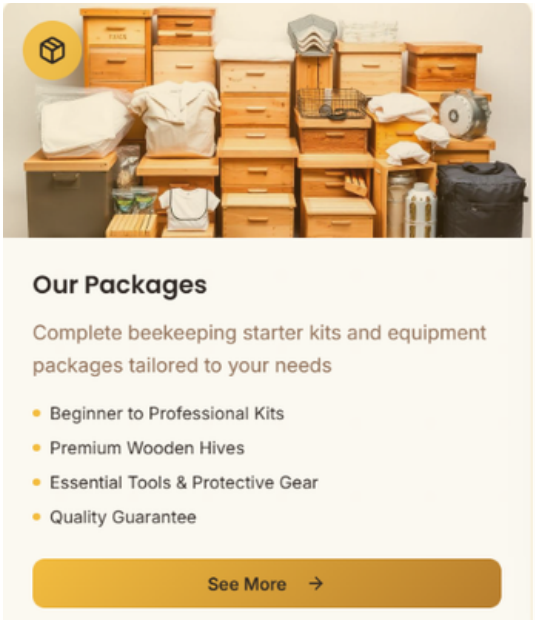
## Training and Testing Labs

Interested farmers are connected to their nearby training centres for training and information is dissemenated among them.

# Services Offered

## Our Packages

- Beekeeping equipment are provided to beekeepers according to their needs.
- Prices are compared and best deals are handed to farmers.
- equipment reached directly to farmers doorstep.




## How we operate ?

### Online

- First, farmer registers in our website by using one time QR available in catalogue or buy from the store.
- After logging back in now the farmer has access to us.
- Farmer can find eqipment according to their needs and order from us.
- We make sure the items are directly delivered to the farmers.

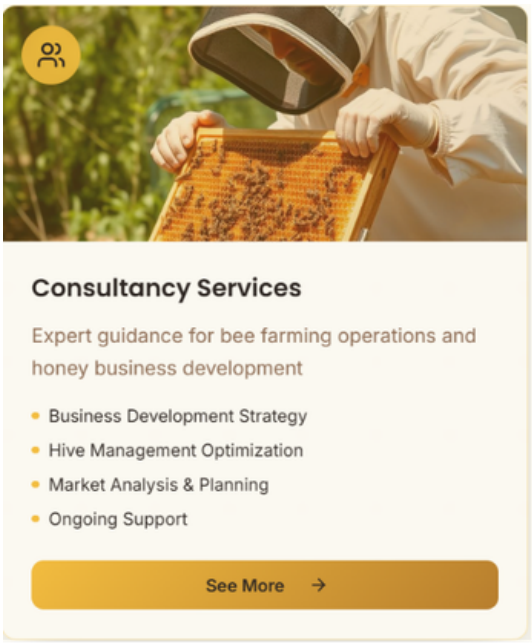
### Offline

- The farmer connects with Hive to Harvest team.
- Now instead of ordering directly, farmer ask us to select for them.
- The team carefully selects items required according to their needs.
- Comparison is made among prices, the team orders on behalf of farmer and the best deal is handed over to them.

# Services Offered

## Consultancy services

- Consultancy services is provided to farmers.
- Starting from setup, management and marketing of their produce.
- Farmers are connected to their nearby KVK's and SAU's.
- Personal consultants are also provided to attend them anytime to resolve their issues.




## How we operate ?

### Online

- First, farmer registers in our website by using one time QR available in catalogue or buy from the store.
- After logging back in now the farmer has access to all the information of nearby KVK's and SAU's along with contact information and access to our private consultant.
- Farmer gets help in apiary setup, linkage to markets for their products and regarding diseases and pest management in hives.

### Offline

- The farmer connects with Hive to Harvest team.
- The farmer now contacts our personal consultant about their issues.
- The consultant according to their issues either schedules a visit with them or if resolvable through call do the needful.
- Farmer can avail the personal consultant services as long as their subscription remains.

# Services Offered

## Training & Testing Labs

- Farmers who wish to take training for beekeeping are connected to their nearby training centres.
- If a farmer wishes to test their honey they are also provided information and connected to their nearby testing labs.




## How we operate ?

### Online

- First, farmer registers in our website by using one time QR available in catalogue or buy from the store.
- After logging back in now the farmer has access to all the information of nearby training centres and testing labs.
- Farmer connects to us and trainings are scheduled for farmers in group or alone according to their preference.
- Our team makes sure that the farmers are placed for the training in their nearest centre.

### Offline

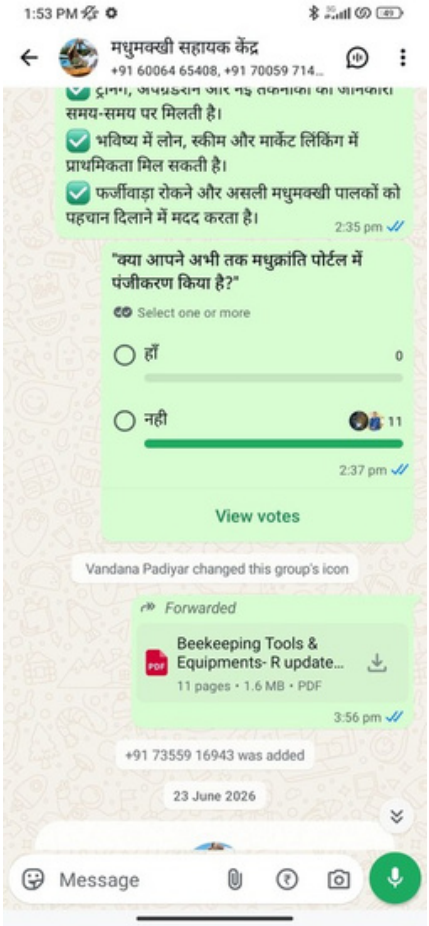
- The farmer connects with Hive to Harvest team.
- If the farmer is alone and not yet connected to any beekeeping community, we first add them to our existing network of beekeepers.
- Now the farmer becomes part of the group and training is scheduled for them.

# Implementation & Progress

| Stakeholder | Description |
|---|---|
| Farmers | More than **200 beekeepers** connected through Hive to Harvest as of now. |
| Equipment Dealers | Network of verified dealers supplying beekeeping equipment.**(8 Suppliers)** |
| Training | More than 100 farmers connected for training in 2-3 groups and successfully completed training. |
| Catalogue | In total 500 catalogues sold so far in kissan mela and to farmers in our area. |
| Personal Consultants | Provided personalized consultancy for hive management, pest and disease management, and other technical issues. |
| Honey Testing Laboratories | Guided farmers to nearby honey testing laboratories for quality analysis and certification support. |

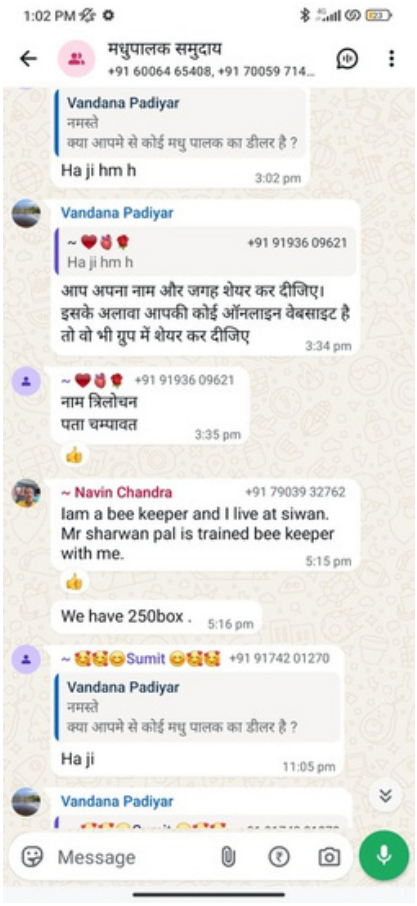
# Implementation & Progress

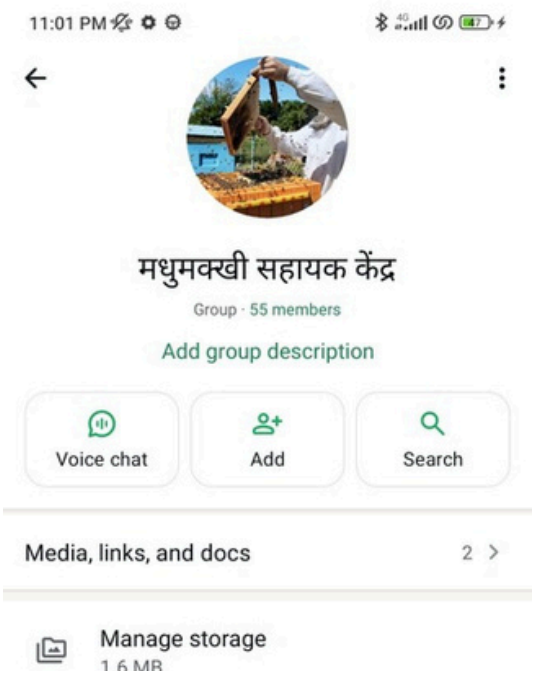
**Farmers/beekeepers -** Few screenshots of the Outreach process

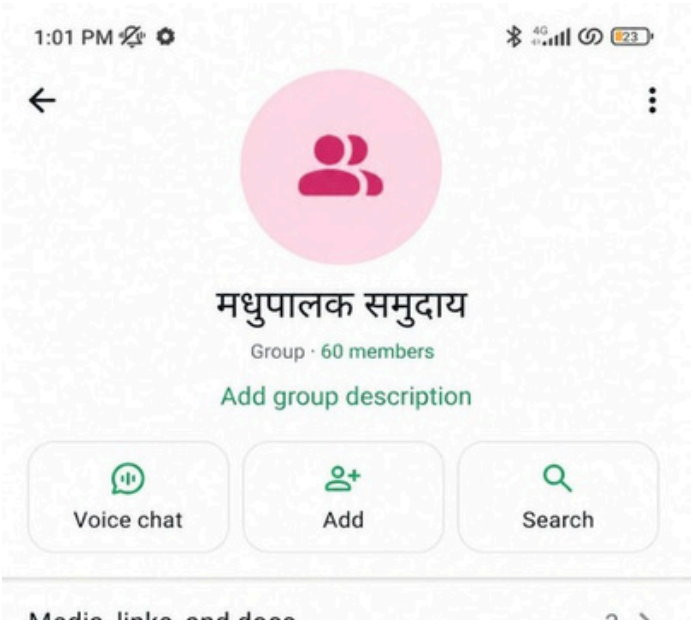




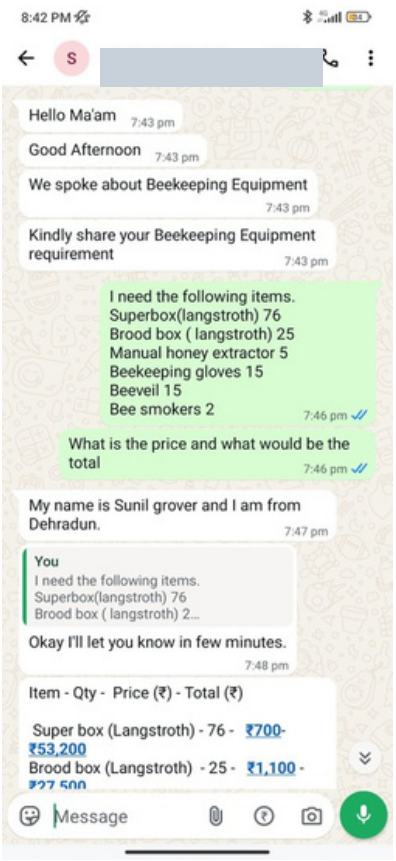


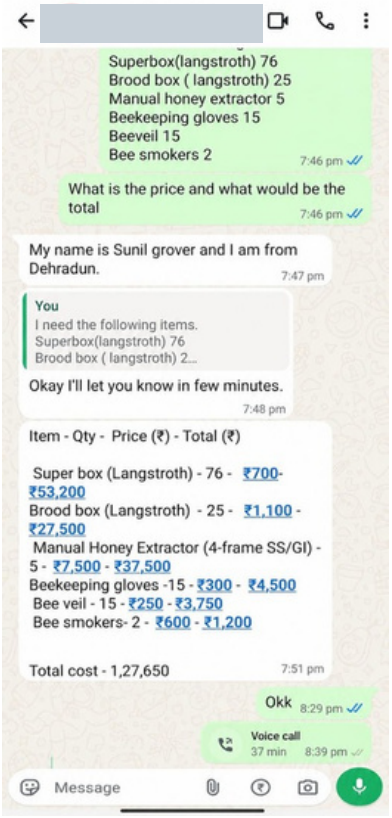


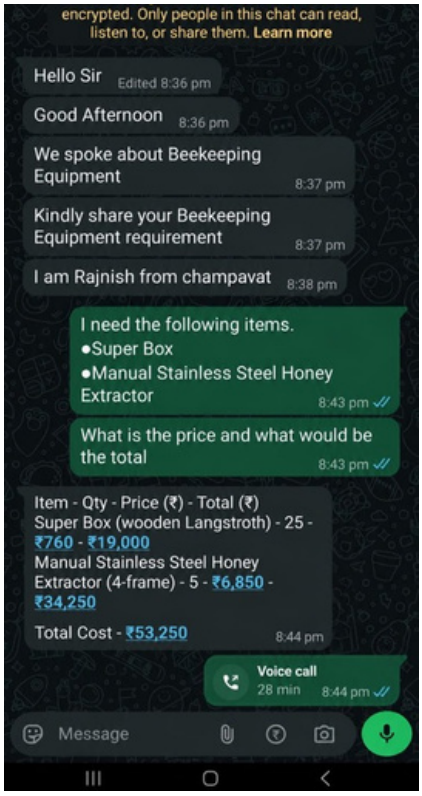



# Implementation & Progress

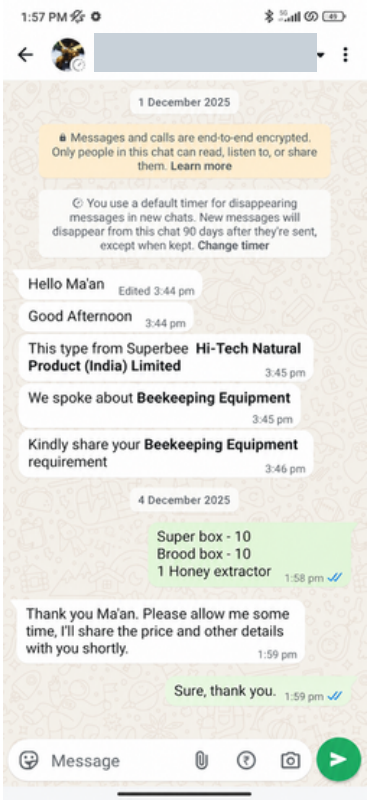


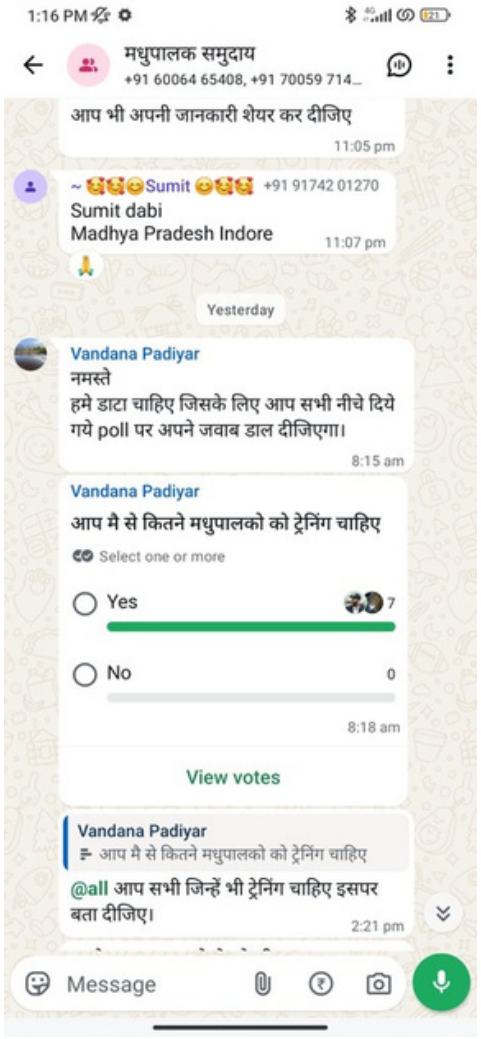


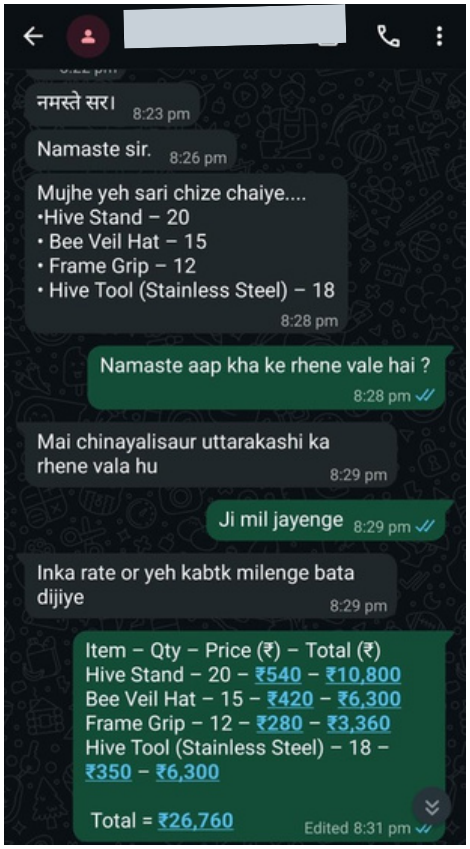




# Implementation & Progress






# Implementation & Progress












# Implementation & Progress



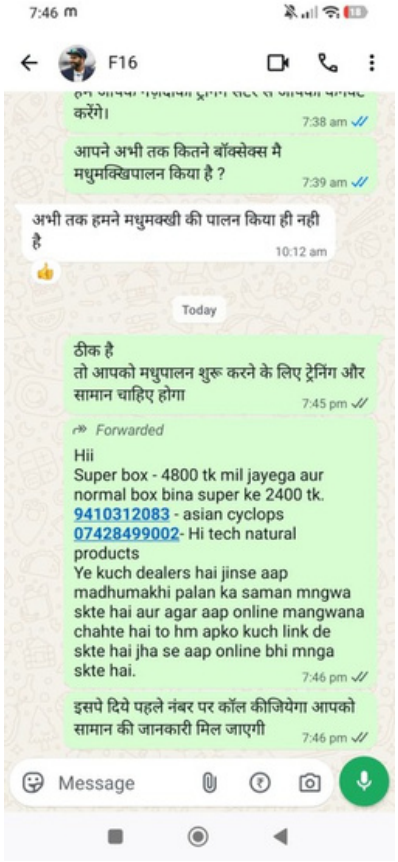






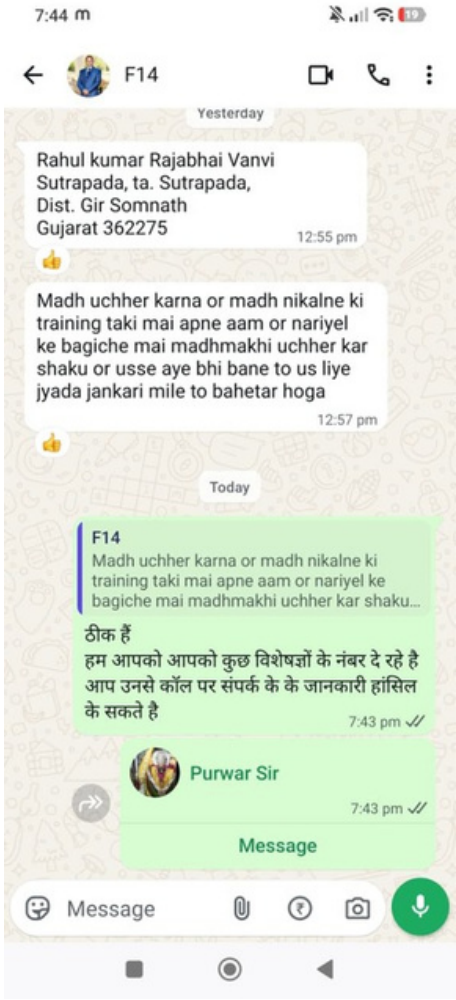


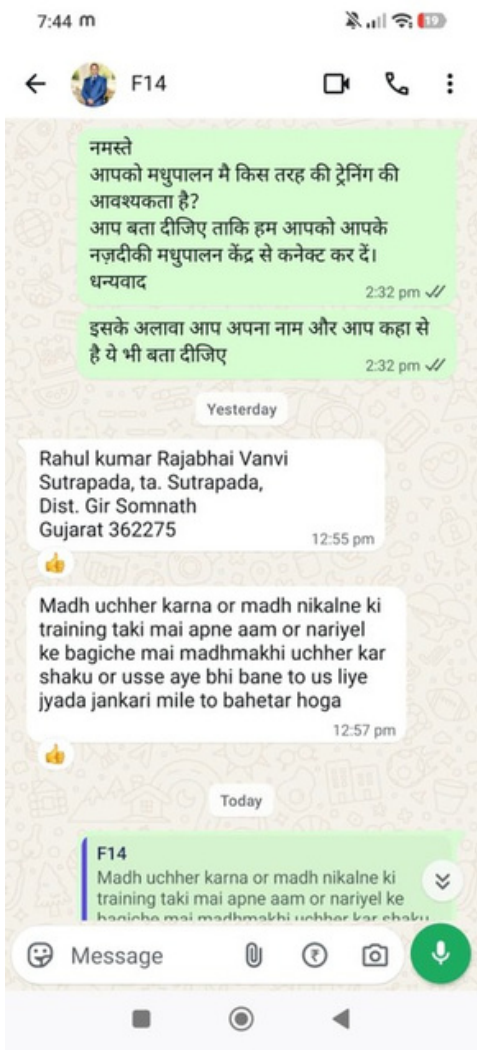

# Implementation & Progress









Catalogue

## Start Your Beekeeping Journey With Us

In this catalogue, you'll find the basic equipment, starter tools, and essential knowledge needed to begin an apiary. And the best part? You can buy all the listed equipment directly from our website through trusted and verified links — so you don't have to search anywhere else.

## Equipments



Queen cage



Queen cell protector



Swarm catching bag



Dummy board



Bottom board

## Scan. Join. Grow.

Step 1 - Copy the link to reach the website or type following url on chrome.
https://hive-to-harvest-site.vercel.app/

step 2 - Go to sign up page.

Step 3 - Fill all the necessary details.

Step 4 - Scan the QR.

Step 5 - You are logged in.

**!** A QR can be used once , single device at a time.

By joining our platform and community, you get exclusive access to:

- Free personal consultancy sessions (Ask anything — setup, equipment, management, problems).
- Step-by-step beekeeping support, from Day 1 to harvesting.
- Help in apiculture training and certification.
- Simplified beekeeping management guides.
- Help in choosing the right equipment and avoiding costly mistakes.
- Updates, tips, and alerts for better colony health.
- A community of learners and beekeepers you can connect with.



# HIVE TO HARVEST

https://hive-to-harvest-site.vercel.app/

https://hive-to-harvest-site.vercel.app/

# Catalogue

## Equipments


Hive box


Stand


Smoker


Queen excluder

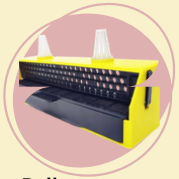

Hive tool


Decapping knife


Bee veil


Gloves


Bee brush


Pollen trap


Honey extractor

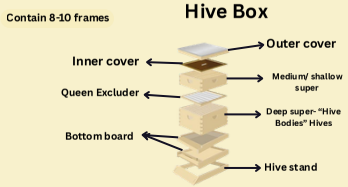

Feeder

## INITIAL SETUP

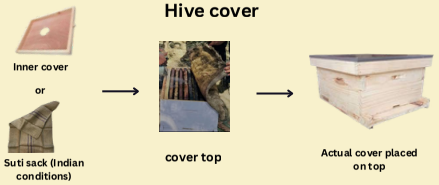
### Hive Placement




### Hive Box




### Hive cover

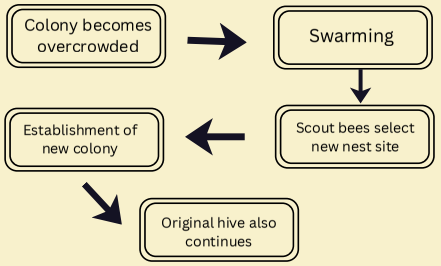



### Feeding



In rainy season, when natural food is scarce plastic feeder with sugar solution is kept inside hive, bees also feed on til, mustard crops, etc.

### Migration (Swarming) Cycle of *Apis mellifera*




### Extraction of honey

1. Inspection of the Hive
2. Smoking the Hive
3. Removing the Frames
4. Uncapping (Removing Wax Layer)
5. Placing Frames in the Extractor
6. Filtering (Straining)
7. Settling
8. Bottling
9. Cleaning the Equipment



## Management of Apis mellifera

**Space and Population**

Provide balanced space for brood, honey, and empty comb.

**Feeding**

Feed sugar syrup during dearth periods. Always maintain reserves of stored honey.

**Colony Health**

Maintain colony health by checking on them every week

**Environment**

Know local nectar plants and blooming seasons. Give shade, protect from rain/cold, and shift colonies if needed.

**Knowledge & Record keeping**

Keep hive notes on queen, brood, honey, and pests.

**Prevent Swarming**

Split colonies if many queen cells are seen, to prevent swarming.

**Pest Management**

i)Wax Moth: Clean hives, expose combs to sun, use Bt spray, sulphur fumigation, or cold storage.

ii).Ants- Keep apiary clean, use water/oil stands, and maintain bowls regularly.

iii)Wasps/Hornets: Trap, reduce hive entrances, destroy nests, and migrate colonies if needed.

iv)Mites
1.Varroa mite: Use formic/oxalic/lactic acid, thymol, and remove drone brood.

2.Tracheal mite: Control with formic acid and. thymol.

3.Reptiles: Keep hives 40–60 cm above ground, apply grease/oil to stands, and clear grass/bushes nearby.

4.Mammals: Bears, raccoons, martens, skunks, rodents damage hives for honey/brood. Protect with fencing, hive stands, reduced entrances, traps, or scare methods.

# Catalogue

## हमारे साथ अपनी मधुमक्खी-पालन यात्रा शुरू करें

इस कैटलॉग में आपको एक मधूपालन केंद्र शुरू करने के लिए आवश्यक बुनियादी उपकरण, प्रारंभिक औज़ार और ज़रूरी जानकारी मिलेगी। और सबसे अच्छी बात? आप यहाँ सूचीबद्ध सभी उपकरणों को हमारी वेबसाइट से विश्वसनीय और प्रमाणित लिंक के माध्यम से सीधे खरीद सकते हैं -- जिससे आपको कहीं और खोजने की आवश्यकता नहीं होगी।

### उपकरण



रानी पिंजरा



रानी कोशिका रक्षक



मधुमक्खी झुंड पकड़ने की थैली



डमी बोर्ड



निचला बोर्ड

## "स्कैन करें. जुड़ें. बढ़ें."

चरण 1 – वेबसाइट तक पहुँचने के लिए लिंक कॉपी करें या क्रोम ब्राउज़र में निम्नलिखित URL टाइप करें।
https://hive-to-harvest-site.vercel.app/

चरण 2 – साइन-अप पेज पर जाएँ।

चरण 3 – सभी आवश्यक विवरण भरें।

चरण 4 – क्यूआर कोड स्कैन करें।

चरण 5 – आप लॉग इन हो चुके हैं।

एक क्यूआर कोड का उपयोग केवल एक बार और एक समय में केवल एक ही डिवाइस पर किया जा सकता है।

**हमारे प्लेटफॉर्म और समुदाय से जूड़कर, आपको विशेष रूप से इन सुविधाओं का लाभ मिलता है:**

- **फ्री पर्सनल कंसल्टेंसी सेशन्स (कुछ भी पूछें — सेटअप, उपकरण, प्रबंधन, समस्याएँ)**

• दिन 1 से लेकर शहद की कटाई तक चरण-दर-चरण मधुमक्खी पालन सहायता ।

• संपर्क विवरण के साथ सत्यापित प्रशिक्षण केंद्र ।

- सरलीकृत मधुमक्खी-पालन प्रबंधन मार्गदर्शिकाएँ ।
- सही उपकरण चुनने में सहायता और महंगी गलतियों से बचाव।
- बेहतर कॉलोनी स्वास्थ्य के लिए अपडेट, सुझाव और अलर्ट।
- मधुमक्खी पालकों का एक समुदाय, जिससे आप जुड़ सकते हैं।




### उपकरण




## प्रारंभिक सेटअप




### शहद निकालने की प्रक्रिया

1. छत्ते का निरीक्षण
2. धुआँ देना
3. फ्रेम निकालना
4. कैपिंग (मोम की परत) हटाना
5. एक्सट्रैक्टर में रखना
6. छानना
7. बैठने देना
8. बोतल में भरना
9. उपकरण साफ करना



### एपिस मेलिफेरा का प्रबंधन

**स्थान और जनसंख्या**

बच्चे, शहद और खाली छत्ते के लिए संतुलित स्थान प्रदान करें

**फीडिंग**

अकाल (सूखाल) की अवधि के दौरान चीनी की चाशनी खिलाएँ। हमेशा संग्रहीत शहद का भंडार बनाए रखें।

**सामुदायिक स्वास्थ्य**

बीमारियों और कीटों के लिए नियमित रूप से निरीक्षण करें। कमज़ोर/आक्रामक रानियों को बदलें और उपकरणों का अच्छी तरह से रखरखाव करें।

**पर्यावरण**

स्थानीय अमृत पौधों और खिलने वाले मौसमों को जानते हैं छाया देना बारिश ठंड से बचाओ और यदि आवश्यक हो तो हिंदी में लिखित कॉलोनियों को शिफ्ट करें

**ज्ञान और अभिलेखन**

छत्ते का निरीक्षण रानी मधुमक्खी , ब्रूड , शहद , और कीटों की स्थिति पर

**झुंड को रोकें**

यदि छत्ते में कई रानी कोशिकाएं दिखाई दें, तो झुंड को रोकने के लिए कॉलोनी को विभाजित कर देना चाहिए।

**कीट प्रबंधन**

1. मोम पतंगा (Wax Moth) नियंत्रण के उपाय: छत्ते साफ रखें ,कंघियों को धूप दिखाएं,बीटी स्प्रे का उपयोग करें,सल्फर धूमन,कोल्ड स्टोरेज।

2. चींटियों से बचाव : मधुमक्खी पालन गृह को साफ रखें ,पानी/तेल के स्टैंड का उपयोग करें,कटोरियों का नियमित रखरखाव करें।

3. माइट
- मरेओआ माइट नियंत्रण : फार्मिक एसिड,ऑक्सालिक एसिड,लैक्टिक एसिड ,थाइमोल,ड्रोन ब्रूड निकालना।
- श्वासनली के घुन (Tracheal Mite): का फॉर्मिक एसिड (Formic Acid) और थाइमोल (Thymol) से नियंत्रण करें।

4. सरीसृपों से बचाव के उपाय :छत्तों को ज़मीन से ऊपर रखें,स्टैंड पर ग्रीस/तेल लगाएँ ,आस-पास की घास/झाड़ियाँ साफ़ करें।

5. भालू, स्कूंक, और कुछ कृंतक (चूहे जैसे जानवर) - इन शिकारियों से बचाव के कुछ तरीके: बाड़ लगाना,छत्ता स्टैंड,प्रवेश द्वार छोटा करना,जाल,डराने के तरीके ।

# Future Scope

- Develop a dedicated support system to enable easier access to consultancy services, equipment procurement, training registrations, and service requests.
- Introduce an integrated equipment ordering and delivery system, allowing farmers to place orders directly through the platform and connecting it to an e-store.
- Build a comprehensive digital knowledge repository containing best management practices, seasonal advisory, training materials, and educational resources for beekeepers.
- Establish collaborations with universities, research institutions, government agencies, and beekeeping organizations to strengthen knowledge exchange and improve service delivery.
- Create a community forum where beekeepers can share experiences, discuss challenges, and learn from experts and fellow practitioners.
- Expand personalized consultancy services by onboarding additional subject experts to provide timely technical guidance for colony management and other beekeeping operations.
- Incorporate multilingual support to improve accessibility for beekeepers from different linguistic regions of the country.
- Strengthen the service network to ensure faster coordination between farmers and nearby equipment dealers, training centres, consultants, and other stakeholders.
- Scale Hive to Harvest into a nationwide integrated apiculture support ecosystem.